

CLOUD COMPUTING

LAB TEST MASTER STUDY NOTE

Consolidated from the supplied past papers, Q&A sheets and scanned MCQ collections

Study focus: AWS networking, EC2, scaling, storage, databases, monitoring, security, HA/DR, cloud-native concepts and IaC

Important source note Some photographed MCQ pages contain student selections or old AWS wording. This guide prioritizes repeated concepts and the explanatory Q&A material rather than blindly copying every marked option.

Prepared as a rapid revision guide

How to Use This Note

Do not study every topic with equal effort. The past-paper pattern is heavily concentrated around a small group of AWS concepts. Start with the high-frequency areas, then use the later sections for theory and one-off questions.

Priority	Topic	Why it matters
★★★★★	VPC, subnets, routing, CIDR, SG/NACL	Most repeated technical area
★★★★★	Auto Scaling, ELB, CloudWatch	Large number of MCQs and scenarios
★★★★☆	EC2, AMI, SSH, EBS, pricing	Repeated practical questions
★★★★☆	S3, RDS, DynamoDB, Redshift, ElastiCache	Many service-selection questions
★★★☆☆	Cloud models and deployment models	Easy marks and repeated fundamentals
★★★☆☆	High Availability, DR, Business Continuity	Dedicated MCQ set
★★★☆☆	IAM, encryption, CloudTrail	Frequent security questions
★★☆☆☆	CloudFormation and DevOps tools	Fill-in-the-blank and scenario questions
★★☆☆☆	Docker, microservices, serverless	Appears in newer papers
★☆☆☆☆	Migration and Well-Architected	More theory-heavy

Major Sections

- 1. Cloud fundamentals and service/deployment models
- 2. AWS global infrastructure
- 3. VPC, CIDR, subnets, routing and gateways
- 4. Security Groups, NACLs and network connectivity
- 5. EC2, AMI, SSH, storage and pricing
- 6. Auto Scaling and Elastic Load Balancing
- 7. Monitoring, logging and notifications
- 8. S3, EBS, CloudFront and storage decisions
- 9. RDS, DynamoDB, Redshift, ElastiCache and Kinesis
- 10. IAM, encryption and shared responsibility
- 11. High availability, business continuity and disaster recovery
- 12. CloudFormation, DevOps, Lambda, Docker and microservices
- 13. Migration and Well-Architected Framework
- 14. Scenario map, trap tables and final cram sheet

1. Cloud Computing Fundamentals

1.1 What is Cloud Computing?

Cloud computing is the delivery of computing capabilities such as servers, storage, networking, databases and applications over a network on demand. The past papers repeatedly contrast this with purchasing and maintaining fixed on-premises infrastructure.

Exam idea Traditional IT is capacity-first: buy, install and maintain. Cloud is demand-first: provision quickly, scale and pay according to use.

1.2 Core Characteristics

Characteristic	Meaning
On-demand self-service	Resources can be provisioned when required.
Broad network access	Services are accessible through networks/Internet.
Resource pooling	Provider infrastructure serves multiple customers.
Rapid elasticity	Resources can expand and shrink quickly.
Measured service	Usage is measured and commonly billed pay-as-you-use.
Scalability	The system can handle more workload by adding resources.
Availability	The service remains accessible despite component failures.

1.3 Scalability vs Elasticity

Concept	Best way to remember it	Example
Scalability	Ability to handle increased workload by adding resources.	Business permanently grows from 2 servers to 8.
Elasticity	Resources expand and shrink in response to changing demand.	Traffic spikes in the evening and falls at night.

Common trap Elasticity includes shrinking back down after the demand disappears.

1.4 Service Models: IaaS, PaaS, SaaS

Model	What you receive	What you mainly manage	Typical example
IaaS	Virtual compute, storage and networking	OS, runtime, applications and data	Amazon EC2
PaaS	Managed runtime and development/deployment platform	Application and data	Elastic Beanstalk style platform
SaaS	Complete ready-to-use application	Configuration and business data	Browser-delivered software

Memory IaaS = infrastructure. PaaS = platform. SaaS = finished software.

1.5 Deployment Models

Model	Meaning	Question clue
Public Cloud	Provider-operated cloud shared across customers.	No separate hardware procurement; pay-as-you-use.
Private Cloud	Cloud environment dedicated to one organization.	Only one organization should access/control it.

Model	Meaning	Question clue
Community Cloud	Cloud shared by organizations with common needs.	"A group of organizations".
Hybrid Cloud	Combination of two or more cloud environments.	On-premises/private resources plus public cloud.

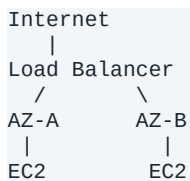
2. AWS Global Infrastructure

2.1 Region, Availability Zone and Edge Location

Term	Meaning	Exam purpose
Region	A geographic AWS area containing multiple Availability Zones.	Choose geographic placement.
Availability Zone (AZ)	An isolated location inside a Region.	Increase availability by spreading resources.
Edge Location	A distributed location close to end users.	CloudFront content delivery/caching.

Critical rule A subnet belongs to one Availability Zone. A Region contains multiple Availability Zones.

2.2 High-Availability Placement Pattern



The papers repeatedly favor distributing application instances across Availability Zones instead of keeping every instance in one AZ.

3. Amazon VPC - Highest Priority

3.1 VPC Mental Model

Amazon Virtual Private Cloud (VPC) is a logically isolated virtual network in AWS. You control the IP address range, subnets, routes, security controls and network gateways.



3.2 VPC Creation Concepts

- The VPC needs an IP address range expressed in CIDR notation.
- The VPC is created in a Region.
- Subnets are created later inside the VPC CIDR range.
- A main route table is created with a new VPC; subnets and an Internet Gateway are not automatically created just because you created the VPC.

3.3 CIDR Calculation

The papers include subnet-size and usable-address questions. Use the basic formula below.

Total IPv4 addresses = $2^{(32 - \text{prefix length})}$
 Example: /28 -> $2^{(32-28)} = 16$ total addresses

CIDR	Total addresses
/24	256
/25	128
/26	64
/27	32
/28	16

AWS subnet rule For the past-paper /28 style question, 16 total addresses minus 5 AWS-reserved addresses gives 11 usable addresses.

Example: 192.168.0.64/28

Network range: 192.168.0.64 - 192.168.0.79

Past-paper usable host range after AWS reservations: 192.168.0.68 - 192.168.0.78

3.4 Subnets

Public Subnet	Private Subnet
Has a route for Internet-bound traffic through an Internet Gateway.	No direct Internet Gateway route for its instances.
Typical: public-facing load balancer, NAT Gateway.	Typical: application servers, databases, internal services.

Fast recognition If the route table includes 0.0.0.0/0 -> Internet Gateway, the subnet is public.

3.5 Route Tables

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxx

The local route allows communication inside the VPC. Security Groups and Network ACLs still decide whether the traffic is allowed.

3.6 Internet Gateway vs NAT Gateway

Feature	Internet Gateway (IGW)	NAT Gateway
Primary use	Public Internet connectivity for a VPC/public subnet	Outbound Internet access for private-subnet resources
Unsolicited inbound Internet	Can be possible if routing and security allow	Not the normal purpose
Typical placement	Attached to the VPC	Placed in a public subnet
Memory	Public Internet path	Private machines go OUT

```

Private EC2
|
NAT Gateway (public subnet)
|
Internet Gateway
|
Internet
  
```

4. Network Security and Connectivity

4.1 Security Group vs Network ACL

Security Group	Network ACL
Applied at instance/ENI level	Applied at subnet level
Stateful	Stateless
Allow rules	Allow and deny rules
Response traffic for an allowed connection is automatically recognized	Return traffic must also satisfy the opposite-direction rules
Primary instance-level firewall	Additional subnet-level control

Memory SG remembers the connection. NACL forgets the connection.

HTTPS example

For a Security Group, allowing the required inbound HTTPS request is enough for the response path because the Security Group is stateful. For a Network ACL, both directions must be permitted because it is stateless.

4.2 Ports You Should Know

Service	Port
SSH	22
HTTP	80
HTTPS	443
RDP	3389
MySQL	3306
PostgreSQL	5432

Database rule If an EC2 web server needs MySQL on RDS, allow TCP 3306 from the web-server Security Group to the DB Security Group rather than exposing the DB to the whole Internet.

4.3 VPC Peering

- VPC peering connects two VPCs privately.
- CIDR ranges must not overlap.
- Route tables need routes to the peer VPC CIDR through the peering connection.
- Peering is not transitive: A-B and B-C does not automatically create A-C connectivity.
- The supplied Q&A explicitly includes cross-Region peering as supported.

Repeated scenario If the peering connection was created successfully but traffic fails, check the route tables first.

4.4 VPN and Direct Connect

Technology	Purpose
VPN	Encrypted network connection over the Internet between on-premises and AWS.
AWS Direct Connect	Dedicated network connectivity between on-premises and AWS.
Fault-tolerant design clue	Use another connection path, commonly a backup VPN or redundant Direct Connect arrangement.

5. Amazon EC2

5.1 What You Configure When Launching EC2

- AMI
- Instance type
- VPC
- Subnet
- Security Group
- Key pair when required
- Storage
- IAM role
- User data

5.2 AMI

An Amazon Machine Image (AMI) is a reusable machine template used to launch EC2 instances. It captures the operating-system and software configuration needed to create instances with the same starting state.

Auto Scaling link Auto Scaling launch configurations/templates point to an AMI so newly launched instances start from a defined image.

5.3 Connecting to Linux EC2 with SSH

```
ssh -i mykey.pem ec2-user@PUBLIC_IP
```

If you get a "Permission denied (publickey)" or cannot connect, check the username, private key, Security Group port 22, instance state, and network reachability.

5.4 Private IP, Public IP and Elastic IP

Address	Use
Private IP	Internal communication inside the VPC.
Public IP/Public DNS	Internet-reachable addressing when routing/security permit it.
Elastic IP	Persistent public IPv4 address that can be associated with an AWS resource.

5.5 Instance Store vs EBS

Instance Store	Amazon EBS
Ephemeral/temporary host-attached storage	Persistent block storage for EC2
Use only for data that can be lost/recreated	Suitable for filesystems, databases and logs
Data persistence is tied to instance lifecycle/host behavior	Can persist independently of simple instance stop/reboot scenarios
Not the default answer when durable disk is required	Can be encrypted and performance is discussed in IOPS

5.6 EC2 Pricing Models

Model	Best fit
On-Demand	Unpredictable workloads; no long-term commitment.

Model	Best fit
Reserved	Steady/predictable usage where commitment can reduce cost.
Spot	Interruptible, fault-tolerant batch or processing workloads.
Dedicated	Dedicated hardware/isolation requirements.

Spot clue If work can resume after interruption and the requirement emphasizes lowest cost for temporary processing, Spot is the key option.

6. Auto Scaling and Elastic Load Balancing

6.1 Auto Scaling Group Basics

Minimum capacity <= Desired capacity <= Maximum capacity

An Auto Scaling Group maintains a requested number of EC2 instances and changes capacity when policies or manual actions require it.

Setting	Meaning
Minimum	Lowest number of instances the group should maintain.
Desired	Target number of instances right now.
Maximum	Highest number of instances the group may scale to.

Repeated MCQ For manual scaling, change the Desired Capacity.

6.2 Auto Scaling Launch Inputs

- AMI
- Instance type
- Security configuration
- Storage and optional key pair
- User data scripts when you need software/configuration at launch

Preinstalled software Two common past-paper answers are: use User Data to install/configure software at boot, or build a "golden" AMI and launch from it.

6.3 Scaling Policies

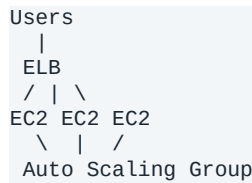
Policy	When to use	Example clue
Dynamic Scaling	Scale using live metrics.	CPU or request count crosses a threshold.
Target Tracking	Maintain a target metric level.	Keep average CPU near a target percentage.
Scheduled Scaling	Known demand at a known time.	Promotion starts next Friday at 6 PM.
Step Scaling	Different scaling amount based on severity.	Moderate alarm +1; severe alarm +4.
Cooldown/Stabilization	Prevent rapid repeated scale actions.	System scales up and down many times in one hour.

6.4 Availability-Zone Behavior

Past-paper rule When Auto Scaling needs another instance, it tries to balance enabled Availability Zones and favors the AZ with fewer instances.

6.5 Elastic Load Balancing (ELB)

ELB distributes incoming traffic among backend targets. The papers connect ELB with high availability, health checks and Auto Scaling.



ELB responsibility	Auto Scaling responsibility
Distribute traffic among healthy targets.	Add/remove instances according to capacity policy.
Perform health checks.	Maintain min/desired/max capacity.
Improve availability across targets/AZs.	Replace unhealthy instances when configured.

6.6 Health Checks and Connection Draining

- Health checks can use HTTP/HTTPS or TCP-style connectivity checks depending on the load-balancing setup.
- An unhealthy target should stop receiving new requests.
- Connection draining allows in-flight requests to complete while an instance is being removed.
- If Auto Scaling incorrectly marks a known-good instance unhealthy, the supplied Q&A mentions manually restoring its health status using the Auto Scaling health command/API.

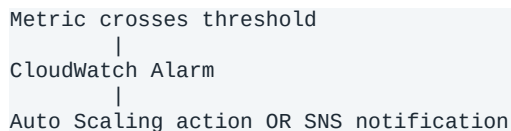
7. Monitoring, Logging and Notifications

7.1 CloudWatch vs CloudTrail

Amazon CloudWatch	AWS CloudTrail
Metrics, alarms and operational monitoring	API activity and audit trail
CPU, latency, request count, network metrics	Who/what called an AWS API
CloudWatch Logs stores/analyzes log data	Useful for governance and security investigation

Memory CloudWatch = watch system behavior. CloudTrail = trail of API actions.

7.2 CloudWatch Patterns



- CloudWatch Logs is the repeated answer for storing and analyzing EC2/system logs.
- Custom metrics can be submitted through CLI/API; the past papers mention put-metric-data.
- CloudWatch is also used to monitor RDS, Redshift and Lambda metrics/logs in the supplied questions.

7.3 SNS

Amazon Simple Notification Service is used for notifications. The past-paper examples include email and SMS endpoints.

CloudWatch Alarm -> SNS -> Email / SMS

8. Storage and Content Delivery

8.1 Amazon S3

Amazon S3 is object storage. Data is stored as objects in buckets and accessed through object-storage APIs rather than as ordinary attached block devices.

- Good for static assets, files, backups, logs, media and data-lake objects.
- Not the normal answer for an EC2 boot disk or database block device.
- The papers test that storage does not need to be preallocated in advance.
- S3 access can be controlled using IAM, bucket policies, ACLs and pre-signed URLs depending on the use case.

8.2 S3 Encryption

Protection goal	Past-paper mechanisms/examples
Encryption at rest	Server-side encryption with KMS-managed keys; server-side encryption with customer-provided keys; client-side encryption.
Encryption in transit	SSL/TLS protects data while moving over the network.

8.3 S3 vs EBS vs Glacier

Service	Storage type	Best use
EBS	Block	EC2 disk, filesystems, transactional storage
S3	Object	Files, backups, static assets, data-lake objects
S3 Glacier / archival class	Archive	Long-term infrequent retrieval

8.4 CloudFront

Amazon CloudFront is AWS content delivery/CDN service. It caches and serves content from edge locations to reduce latency for globally distributed users.

Origin -> CloudFront -> Edge Locations -> Users

9. Databases and Data Services

9.1 Service Selection

Service	Best match in the past papers
Amazon RDS	Relational OLTP / managed relational database
Amazon DynamoDB	Highly scalable NoSQL key-value/document workloads
Amazon Redshift	Data warehouse / OLAP analytics
Amazon ElastiCache	Managed in-memory caching
Amazon Kinesis	Streaming/near-real-time data ingestion and processing

Fast keyword map OLTP -> RDS. OLAP/data warehouse -> Redshift. NoSQL -> DynamoDB. Cache -> ElastiCache. Stream -> Kinesis.

9.2 RDS Multi-AZ vs Read Replica

Multi-AZ	Read Replica
High availability/failover pattern	Read scaling/reporting pattern
Standby copy in another AZ	Additional readable copy/copies
Think: resilience	Think: more read capacity

Repeated MCQ For "Which RDS feature provides high availability?" choose Multi-AZ deployment.

9.3 RDS Security

- Place databases in private subnets when public exposure is unnecessary.
- Use Security Groups to allow only the required application source and DB port.
- For MySQL, the recurring port is 3306.
- The old Q&A notes that RDS encryption must be considered at creation rather than assuming an already-running unencrypted instance can simply be toggled in place; follow your current lab material for current service behavior.

10. IAM, Encryption and Shared Responsibility

10.1 IAM

AWS Identity and Access Management controls identities and permissions. The core past-paper principle is least privilege: grant only the actions and resources required.

IAM concept	Purpose
User	Long-term AWS identity for a person/application when appropriate.
Group	Collection of users for common permissions.
Role	Assumable identity that provides temporary permissions.
Policy	Permission document defining allowed/denied actions/resources.

Best practice Use roles for AWS workloads instead of hard-coding access keys where possible. Do not expose access keys in source repositories.

10.2 Shared Responsibility Model

Provider / AWS	Customer
Physical data centers and physical hardware	Data and application security
Underlying cloud infrastructure	IAM permissions and account configuration
Core managed-service infrastructure	Guest OS management for IaaS where applicable
Platform security of the cloud	Network rules, encryption choices and secure application design

Memory AWS = security OF the cloud. Customer = security IN the cloud. The exact split changes with IaaS/PaaS/SaaS.

10.3 Data Security Checklist

- Authentication and authorization
- Least-privilege IAM
- Encryption at rest
- Encryption in transit
- Private database/network placement

- Security Groups and NACLs
- Logging and monitoring
- Backups and recovery planning

11. High Availability, Business Continuity and Disaster Recovery

11.1 High Availability

High availability means designing a service so it can keep operating when individual components fail. The dedicated MCQ set repeatedly emphasizes redundancy, monitoring and failover.

HA building block	Purpose
Redundancy	Avoid a single copy/component.
Monitoring	Detect failure quickly.
Failover	Move service to a healthy component.
Load Balancing	Distribute traffic across multiple backends.
Clustering	Use multiple cooperating nodes.
Auto Scaling	Maintain/adjust healthy capacity.
Backup and Recovery	Recover data/services after loss.

11.2 Fault Tolerance vs High Availability

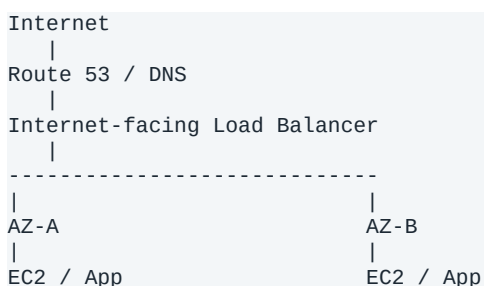
High Availability	Fault Tolerance
Minimizes downtime and maintains service availability.	Continues operation despite component failure with stronger redundancy.
May still involve brief failover.	Targets minimal or no service interruption.
Common AWS pattern: multi-AZ + ELB + ASG.	Usually more expensive and redundant.

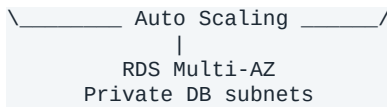
11.3 Business Continuity vs Disaster Recovery

Business Continuity (BC)	Disaster Recovery (DR)
Keep critical business operations running at acceptable levels.	Restore technology/data after major disruption.
Broader organizational continuity planning.	Technology-focused recovery planning.
Think: keep the business functioning.	Think: recover systems and data.

DR plan elements from the papers Risk evaluation, business-critical asset identification and backup strategy.

11.4 Standard Highly Available AWS Architecture





- Private instances that need outbound Internet access can use a NAT Gateway.
- On-premises access can use VPN or Direct Connect depending on requirements.
- CloudWatch monitors; CloudTrail audits API activity; IAM controls permissions.

12. Cloud-Native, DevOps and Infrastructure as Code

12.1 CloudFormation

AWS CloudFormation is Infrastructure as Code. A template describes AWS resources; CloudFormation creates and manages them as a stack.

Template -> CloudFormation -> Stack -> AWS Resources

Old-paper template detail The supplied Q&A identifies Resources as the mandatory core section, while sections such as Parameters and Outputs are optional.

12.2 AWS Developer/DevOps Tool Mapping

Service	Purpose
CodeCommit	Source-code repository
CodeBuild	Compile/build/test/package
CodeDeploy	Deployment automation
CodePipeline	CI/CD workflow orchestration
CodeStar	Integrated development-project tooling
Cloud9	Browser-based development environment
CloudFormation	Infrastructure as Code

12.3 Lambda and FaaS

AWS Lambda is the repeated serverless/FaaS example: run functions without managing servers, respond to events and scale execution automatically.

Event -> Lambda function -> Output / action

12.4 Monolith vs Microservices

Monolith	Microservices
Application packaged/deployed largely as one unit.	Application split into smaller independently deployable services.
Simpler initial deployment.	Independent scaling and deployment.
Changes can affect the whole application.	More distributed-system/network complexity.

Independent microservices commonly communicate through REST/HTTP APIs, gRPC or message/event systems.

12.5 Docker

Docker packages applications and dependencies into containers. An image is a reusable template; a container is a running instance of an image.

```

docker images
docker ps
docker ps -a
docker run IMAGE
docker stop CONTAINER
  
```

```
docker rm CONTAINER
docker rmi IMAGE
docker pull IMAGE
```

Repeated command The scanned paper explicitly asks for the command that lists Docker images: docker images.

12.6 Spring Boot Actuator and Blue-Green Deployment

- Spring Boot Actuator exposes operational health/metrics information for a running Spring Boot application.
- Blue-Green deployment keeps two production environments: one live and one candidate. Traffic is switched after validation, reducing deployment risk and downtime.

12.7 SDN and NFV

SDN	NFV
Software Defined Networking: programmable/abstracted network control.	Network Functions Virtualization: network functions implemented as software.
Common conceptual layers: application, control, infrastructure/physical.	Examples: virtual firewall, router or load-balancer function.

13. Cloud Migration and Well-Architected Framework

13.1 Migration Strategies

Strategy	Meaning	Typical clue
Rehost / Lift-and-Shift	Move the application with minimal change.	Fastest move; limited redesign.
Replatform	Make limited cloud optimizations while retaining core architecture.	Example: move DB to a managed database service.
Rearchitect / Refactor	Redesign substantially for cloud-native patterns.	Monolith -> microservices/serverless.

13.2 Well-Architected Framework Pillars

Pillar	Core question
Operational Excellence	Can we operate and improve the system effectively?
Security	Are data, identities and systems protected?
Reliability	Can the workload recover and continue functioning?
Performance Efficiency	Are resources used efficiently for the workload?
Cost Optimization	Are we avoiding unnecessary cost?
Sustainability	Are we minimizing unnecessary resource/environmental impact?

Memory Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.

14. Scenario Keyword Map

Question wording / requirement	Best first answer to consider
API activity / audit / who changed something	CloudTrail
CPU, latency, request count, metrics	CloudWatch

Question wording / requirement	Best first answer to consider
EC2/system log files	CloudWatch Logs
Send email/SMS from alarm	SNS
Distribute web traffic	ELB
Automatically add/remove EC2 instances	Auto Scaling
Known traffic increase at a known time	Scheduled Scaling
Keep a metric around a target	Target Tracking
Private instance needs outbound Internet	NAT Gateway
Public subnet Internet path	Internet Gateway
Private connectivity between VPCs	VPC Peering
On-premises encrypted Internet connection	VPN
Dedicated on-premises to AWS connection	Direct Connect
EC2 block disk	EBS
Object/file storage	S3
Long-term archive	S3 Glacier / archival storage
Global CDN	CloudFront
Relational OLTP	RDS
Relational DB high availability	RDS Multi-AZ
NoSQL key-value/document	DynamoDB
Data warehouse / OLAP	Redshift
In-memory cache	ElastiCache
Streaming ingestion / near-real-time stream	Kinesis
DNS	Route 53
Infrastructure as Code	CloudFormation
Run function without managing servers	Lambda
Source repository	CodeCommit
Build/test/package	CodeBuild

15. High-Yield Trap Tables

15.1 Security Group vs NACL

Statement	Correct?
Security Group is stateful	YES
Network ACL is stateful	NO
Network ACL is stateless	YES

Statement	Correct?
Security Group is the subnet-level control	NO
Network ACL is applied at subnet level	YES
Security Group automatically recognizes return traffic for allowed stateful connections	YES
Network ACL may require rules in both directions	YES
Network ACL supports allow and deny rules	YES

15.2 Storage and Database Traps

Statement	Correct?
S3 is block storage	NO
EBS is block storage	YES
EBS can be attached to EC2	YES
RDS is a managed relational database service	YES
Redshift is the normal OLTP answer	NO
Redshift is for data-warehouse/analytical workloads	YES
DynamoDB is a relational database	NO
RDS Multi-AZ improves availability	YES
Glacier is for frequent low-latency database transactions	NO

15.3 Networking Traps

Situation	What to remember
Peering exists but traffic fails	Check route tables and security.
Peer CIDRs overlap	Peering cannot be used with overlapping address spaces.
A peers with B; B peers with C	A does not automatically peer with C.
Subnet route 0.0.0.0/0 -> IGW	Public subnet.
Private subnet needs outbound Internet	NAT Gateway.
Subnet spans multiple AZs	Wrong: a subnet is tied to one AZ.
Database open to 0.0.0.0/0 on 3306	Usually insecure; allow only required application source.

15.4 CloudWatch vs CloudTrail

Requirement	Service
CPU utilization	CloudWatch
ELB latency	CloudWatch
Metric alarm	CloudWatch

Requirement	Service
EC2 logs	CloudWatch Logs
Lambda logs	CloudWatch Logs
Who called an AWS API	CloudTrail
Security/API audit history	CloudTrail

16. Final Cram Sheet - Memorize These First

1. Security Group vs NACL: stateful vs stateless.
2. Public vs private subnet; IGW vs NAT Gateway.
3. CIDR address counts and AWS subnet reserved-address idea.
4. VPC peering: non-overlapping CIDRs, routes required, no transitive peering.
5. EC2 + AMI + key pair + EBS.
6. Auto Scaling: min, desired, max; manual scaling changes desired capacity.
7. Scheduled vs dynamic vs target-tracking scaling.
8. ELB + health checks + Auto Scaling relationship.
9. CloudWatch vs CloudTrail vs SNS.
10. S3 vs EBS vs Glacier vs CloudFront.
11. RDS vs DynamoDB vs Redshift vs ElastiCache vs Kinesis.
12. RDS Multi-AZ = HA; read replica = read scaling.
13. Region vs AZ vs Edge Location.
14. IAM and least privilege.
15. IaaS vs PaaS vs SaaS.
16. Public vs private vs community vs hybrid cloud.
17. CloudFormation, Lambda, Docker and microservices.
18. High availability, business continuity and disaster recovery.

Exam strategy When a scenario question looks long, first extract three things: (1) what resource is being discussed, (2) whether the requirement is availability, security, performance, cost or connectivity, and (3) the keyword that maps to one AWS service. Then eliminate options that solve a different requirement.

17. One-Off / Low-Priority Items Seen in the Supplied Papers

A few pages in the merged scans include isolated terms that are not central to the AWS sections. Keep them low priority unless your lecturer specifically covered them.

- Knowledge Codification: making institutional knowledge visible, accessible and usable for decision making.
- Concept Mapping: a structured process that creates an interpretable pictorial view of ideas/concepts and their relationships.

Old-paper caution One scan includes source-specific or dated wording such as a question about the "latest default AWS Region." Treat such items as historical paper artifacts rather than stable cloud concepts unless your current lab material repeats them.

18. 60-Second Service Lookup

Need	Service
Virtual machine	EC2
Block storage	EBS
Object storage	S3

Need	Service
Archive	Glacier
CDN	CloudFront
Relational DB	RDS
NoSQL	DynamoDB
Data warehouse	Redshift
Cache	ElastiCache
Streaming	Kinesis
DNS	Route 53
Metrics/alarms	CloudWatch
Logs	CloudWatch Logs
API audit	CloudTrail
Notifications	SNS
Scale EC2	Auto Scaling
Distribute traffic	ELB
Virtual network	VPC
Public Internet path	Internet Gateway
Private outbound Internet	NAT Gateway
Private VPC-to-VPC link	VPC Peering
On-premises encrypted tunnel	VPN
Dedicated on-premises link	Direct Connect
Identity and permissions	IAM
Infrastructure as Code	CloudFormation
Serverless function	Lambda

END OF MASTER NOTE